2.20. *QR*-DECOMPOSITION OF MATRICES

Using the *QR*-decomposition of matrices is an effective approach to solving the full eigenvalue problem for non-symmetric matrices $A_{n\times n}$; it is based on reducing matrices to similar ones having a triangular or quasi-triangular form and allows finding both real and complex eigenvalues. The *QR*-algorithm is based on the representation of the matrix in the form $A=QR$, where Q is the orthogonal matrix ($Q^{-1}=Q^T$), and R is the upper triangular one. This decomposition exists for any square matrix.

One of the possible approaches to the construction of *QR*-decomposition is the use of a Householder transformation that allows turning a group of subdiagonal elements of the matrix column to zero. A Householder transformation is carried out using a Householder matrix, which has the following form:

$$H=E-\frac{2}{v^Tv}vv^T,$$

where v is an arbitrary nonzero column vector, E is the unity matrix, vv^T is the square matrix of the same dimension.

It is easy to make sure that any matrix of this kind is symmetric and orthogonal. At the same time, the arbitrariness in the choice of the vector v makes it possible to construct a matrix that meets some additional requirements.

Let's consider a case when it is necessary to zero out all the elements of some vector except for the first one, i.e. to construct such a Householder matrix that $\tilde{b}=Hb$, $b=(b_1,b_2,...,b_n)^T$, $\tilde{b}=(\tilde{b}_1,0,...,0)^T$.

Then the vector v is determined as follows: $v=b+\text{sign}(b_1)\|b\|_2 e_1$, where $\|b\|_2=\left(\sum_i b_i^2\right)^{1/2}$ is the Euclidean norm of the vector b, and $e_1=(1,0,...,0)^T$.

Applying the described procedure in order to zero out the subdiagonal elements for each of the columns of the original matrix, we can obtain its *QR*-decomposition in a fixed number of steps $n-1$.

Indeed, let's suppose $A_0 = A$ and, at the first step, construct a Householder transformation H_1 ($A_1 = H_1 A_0$), which transforms the matrix A_0 into the matrix A_1 with zero elements in the first column under the main diagonal:

$$A_0 = \begin{pmatrix} a_{11}^0 & a_{12}^0 & \dots & a_{1n}^0 \\ a_{21}^0 & a_{22}^0 & \dots & a_{2n}^0 \\ \dots & \dots & \dots & \dots \\ a_{n1}^0 & a_{n2}^0 & \dots & a_{nn}^0 \end{pmatrix} \xrightarrow{H_1} A_1 = \begin{pmatrix} a_{11}^1 & a_{12}^1 & \dots & a_{1n}^1 \\ 0 & a_{22}^1 & \dots & a_{2n}^1 \\ \dots & \dots & \dots & \dots \\ 0 & a_{n2}^1 & \dots & a_{nn}^1 \end{pmatrix}.$$

It is clear that the Householder matrix H_1 must be determined by the first column of the matrix A_0, i.e. the vector $(a_{11}^0, a_{21}^0, ..., a_{n1}^0)^T$ of the dimension *n* is taken as the vector *b* in the expression. Then, the components of the vector *v* are calculated as

$$v_1^1 = a_{11}^0 + \text{sign}(a_{11}^0)\left(\sum_{j=1}^{n}(a_{j1}^0)^2\right)^{1/2},\ v_i^1 = a_{i1}^0,\ i = 2,\dots,n, \text{ and } H_1 = E - 2\frac{v^1 v^{1T}}{v^{1T} v^1}.$$

At the second step of the process under consideration, a Householder transformation H_2 ($A_2 = H_2 A_1$) is constructed, which zeroes out the elements of the second column of the matrix A_1 located below the main diagonal. Taking the vector $(a_{22}^1, a_{32}^1, ..., a_{n2}^1)^T$ of the dimension $n-1$ as the vector *b*, we obtain the following expressions for the components of the vector *v*:

$$v_1^2 = 0,\ v_2^2 = a_{22}^1 + \text{sign}(a_{22}^1)\left(\sum_{j=2}^{n}(a_{j2}^1)^2\right)^{1/2},\ v_i^2 = a_{i2}^1,\ i = 3,\dots,n, \text{ and } H_2 = E - 2\frac{v^2 v^{2T}}{v^{2T} v^2}.$$

If we repeat this process exactly $n-1$ times for $p = 1,\dots,n-1$ as

$$v_i^p = 0,\ \ i = 1,\dots,p-1,\ \ v_p^p = a_{pp}^{p-1} + \text{sign}(a_{pp}^{p-1})\left(\sum_{j=p}^{n}(a_{jp}^{p-1})^2\right)^{1/2},\ \ v_i^p = a_{ip}^{p-1},\ \ i = p+1,\dots,n,$$

and $H_p = E - 2\frac{v^p v^{pT}}{v^{pT} v^p}$, $A_p = H_p A_{p-1}$ up to $A_{n-1} = H_{n-1} H_{n-2} ... H_1 A$, we shall obtain the desired decomposition $A = QR$, where

$$R = A_{n-1} \text{ and } Q = (H_{n-1} H_{n-2} ... H_1)^{-1} = (H_{n-1} H_{n-2} ... H_1)^T = H_1 H_2 ... H_{n-1}.$$

It should be noted that the process under consideration is in some way similar to the Gaussian method algorithm. The difference lies in the fact that here

the subdiagonal elements of the corresponding column are zeroed out using an orthogonal transformation.

The procedure of *QR*-decomposition is repeatedly used in the iterative *QR*-algorithm for the calculation of eigenvalues of arbitrary matrices.

The following iterative process is constructed:

$$A^{(0)} = A,\ A^{(0)} = Q^{(0)}R^{(0)},\ A^{(1)} = R^{(0)}Q^{(0)},\ \ldots,\ A^{(k)} = Q^{(k)}R^{(k)},\ A^{(k+1)} = R^{(k)}Q^{(k)}.$$

So, each iteration is implemented in two stages. At the first stage, *QR*-decomposition of the matrix $A^{(k)}$ into the product of orthogonal $Q^{(k)}$ and upper triangular $R^{(k)}$ matrices is carried out, and at the second stage, the resulting matrices are multiplied in reverse order.

It is easy to see that the matrices $A^{(k+1)}$ and $A^{(k)}$ are similar. Indeed, given the orthogonality of $Q^{(k)}$ ($Q^{(k)T}Q^{(k)} = E$), we can write:

$$A^{(k+1)} = R^{(k)}Q^{(k)} = E \cdot R^{(k)}Q^{(k)} = Q^{(k)T}Q^{(k)} \cdot R^{(k)}Q^{(k)} = Q^{(k)T}A^{(k)}Q^{(k)}.$$

Accordingly, any of the matrices $A^{(k)}$ is similar to the matrix *A*.

If the matrix A_{nxn} has no multiple eigenvalues, then when $k \to \infty$, the sequence $A^{(k)}$ converges either to an upper triangular matrix if all eigenvalues are real, or to an upper quasi-triangular matrix if there are complex pairwise-conjugate eigenvalues. So, each real eigenvalue will correspond to a matrix column with subdiagonal elements tending to zero, and the eigenvalue itself is taken equal to the diagonal element of this column. Each complex-conjugate pair corresponds to a diagonal block of 2×2 dimension, i.e. the matrix $A^{(k)}$ has a block-diagonal structure. Fundamentally, the elements of these blocks change from iteration to iteration without any apparent regularity, while the complex pairwise-conjugate eigenvalues $\lambda_{1,2}^{(k)}$, determined by each block, tend to converge. This circumstance must be taken into account when forming the criterion for ending the iterative process. If in the course of iterations a complex pairwise-conjugate eigenvalues corresponding to the matrix block formed by the elements of the *j*-th and $(j+1)$-th columns $a_{jj}^{(k)}, a_{j(j+1)}^{(k)}, a_{(j+1)j}^{(k)}, a_{(j+1)(j+1)}^{(k)}$ is traced, then, despite a significant change of these elements in the course of iterations, the eigenvalues corresponding to this

block differ insignificantly starting with some *k*. They are determined from the solution of the quadratic equation

$$(a_{jj}^{(k)} - \lambda^{(k)})(a_{(j+1)(j+1)}^{(k)} - \lambda^{(k)}) = a_{j(j+1)}^{(k)} a_{(j+1)j}^{(k)},$$

$$(\lambda^{(k)})^2 - (a_{jj}^{(k)} + a_{(j+1)(j+1)}^{(k)})\lambda^{(k)} + a_{jj}^{(k)} a_{(j+1)(j+1)}^{(k)} - a_{j(j+1)}^{(k)} a_{(j+1)j}^{(k)} = 0,$$

$$\lambda_{j,j+1}^{(k)} = \frac{1}{2}(a_{jj}^{(k)} + a_{(j+1)(j+1)}^{(k)}) \pm \frac{1}{2}\sqrt{(a_{jj}^{(k)} + a_{(j+1)(j+1)}^{(k)})^2 - 4(a_{jj}^{(k)} a_{(j+1)(j+1)}^{(k)} - a_{j(j+1)}^{(k)} a_{(j+1)j}^{(k)})}.$$

For example, the matrix

$$A^{(k)} = \begin{pmatrix} x & x & x & x & x & x \\ 0 & x & x & x & x & x \\ 0 & x & x & x & x & x \\ 0 & 0 & 0 & x & x & x \\ 0 & 0 & 0 & 0 & x & x \\ 0 & 0 & 0 & 0 & x & x \end{pmatrix}_{6\times 6}$$

has two real eigenvalues $\lambda_1 \approx a_{11}^{(k)}, \lambda_4 \approx a_{44}^{(k)}$ and two pairs of complex-conjugate eigenvalues $\lambda_{2,3}$ and $\lambda_{5,6}$, determined from the equations $(a_{22}^{(k)} - \lambda)(a_{33}^{(k)} - \lambda) = a_{23}^{(k)} a_{32}^{(k)}$ and $(a_{55}^{(k)} - \lambda)(a_{66}^{(k)} - \lambda) = a_{56}^{(k)} a_{65}^{(k)}$.

Since it is impossible to take into account in advance which elements of the matrix A_{nxn} lying on the lower diagonal adjacent to the main one tend to zero when $k \to \infty$ (but there are at least $[(n-1)/2]$ of them, i.e. starting from $n = 3$ there always is at least one such element), the following inequality can be used as the simplest criterion for the termination of iterations:

$$\varepsilon^{(k)} = \left(\sum_{l>m+1}^{n} (a_{lm}^{(k)})^2 + \min_{l=m+1}^{n} (a_{lm}^{(k)})^2 \right)^{1/2} \le \varepsilon, \; n \ge 3.$$

A significant drawback of the *QR*-algorithm for calculating the eigenvalues of matrices is a large number of operations (proportional to n^4) required for the *QR*-decomposition of the matrix at each iteration.

Example. Using the *QR*-decomposition method, calculate the eigenvalues of the matrix $A = \begin{pmatrix} 4 & -1 & -5 \\ 2 & 5 & 8 \\ 1 & -1 & -2 \end{pmatrix} = A^{(0)}$ with an accuracy of $\varepsilon = 0.001$.

Let's note that $\varepsilon^{(0)} = \left(\sum_{l>m+1}^{n} (a_{lm}^{(0)})^2 + \min_{l=m+1}^{n} (a_{lm}^{(0)})^2 \right)^{1/2} = (1^2 + (-1)^2)^{1/2} = \sqrt{2} = 1.412 > \varepsilon$.

Let's start the iterative process. Iteration $k = 1$. $A_0^{(0)} = A^{(0)}$.

Step $p = 1$. Let's find the Householder matrix $H_1^{(0)}$:

$$\sum_{j=1}^{n} (a_{j1}^0)^2 = 4^2 + 2^2 + 1^2 = 21,$$

$$v_1^1 = a_{11}^0 + \text{sign}(a_{11}^0) \left(\sum_{j=1}^{n} (a_{j1}^0)^2 \right)^{1/2} = 4 + 1 \cdot \sqrt{21} = 8.583,$$

$v_i^1 = a_{i1}^0$, $i = 2, \ldots, n$, $v^1 = \begin{pmatrix} 8.583 & 2 & 1 \end{pmatrix}^T$,

$$v^{1T} v^1 = 21 - 4^2 + 8.583^2 = 78.661.$$

$$H_1^{(0)} = E - 2 \frac{v^1 v^{1T}}{v^{1T} v^1} = \begin{pmatrix} -0.873 & -0.436 & -0.218 \\ -0.436 & 0.898 & -0.051 \\ -0.218 & -0.051 & 0.975 \end{pmatrix}.$$

$$A_1^{(0)} = H_1^{(0)} A_0^{(0)} = \begin{pmatrix} -4.583 & -1.091 & 1.309 \\ 0 & 4.979 & 9.470 \\ 0 & -1.011 & -1.265 \end{pmatrix}.$$

Step $p = 2$. Let's find the Householder matrix $H_2^{(0)}$:

$$\sum_{j=2}^{n} (a_{j2}^1)^2 = 4.979^2 + (-1.011)^2 = 25.810,$$

$$v_1^2 = 0,\ v_2^2 = a_{22}^1 + \text{sign}(a_{22}^1) \left(\sum_{j=2}^{n} (a_{j2}^1)^2 \right)^{1/2} = 4.979 + 1 \cdot \sqrt{25.810} = 10.059,$$

$v_i^2 = a_{i2}^1$, $i = 3, \ldots, n$, $v^2 = \begin{pmatrix} 0 & 10.059 & -1.011 \end{pmatrix}^T$,

$$v^{2T} v^2 = 25.810 - 4.979^2 + 10.059^2 = 102.206.$$

$$H_2^{(0)} = E - 2 \frac{v^2 v^{2T}}{v^{2T} v^2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0.980 & 0.199 \\ 0 & 0.199 & 0.980 \end{pmatrix}.$$

$$A_2^{(0)} = H_2^{(0)} A_1^{(0)} = \begin{pmatrix} -4.583 & -1.091 & 1.309 \\ 0 & -5.080 & -9.533 \\ 0 & 0 & 0.644 \end{pmatrix} = R^{(0)}.$$

By doing $n - 1 = 2$ steps for $p = 1, \ldots, n - 1 = 2$, we got the desired decomposition $A^{(0)} = Q^{(0)} R^{(0)}$ of the matrix $A^{(0)}$, where $R^{(0)} = A_{n-1}^{(0)} = A_2^{(0)}$ and

$$Q^{(0)} = H_1^{(0)} H_2^{(0)} ... H_{n-1}^{(0)} = H_1^{(0)} H_2^{(0)} = \begin{pmatrix} -0.873 & 0.384 & -0.301 \\ -0.436 & -0.890 & 0.129 \\ -0.218 & 0.244 & 0.945 \end{pmatrix}.$$

Let's calculate the matrix $A^{(1)}$:

$$A^{(1)} = R^{(0)} Q^{(0)} = \begin{pmatrix} 4.190 & -0.470 & 2.475 \\ 4.297 & 2.201 & -9.663 \\ -0.141 & 0.157 & 0.609 \end{pmatrix}.$$

Let's calculate the error $\varepsilon^{(1)}$:

$$\varepsilon^{(1)} = \left(\sum_{l>m+1}^{n} (a_{lm}^{(1)})^2 + \min_{l=m+1}^{n} (a_{lm}^{(1)})^2 \right)^{1/2} = ((-0.141)^2 + 0.157^2)^{1/2} = 0.211 > \varepsilon.$$

Let's continue the iterative process. Iteration $k = 2$. $A_0^{(1)} = A^{(1)}$.

Step $p = 1$. Let's find the Householder matrix $H_1^{(1)}$:

$$\sum_{j=1}^{n} (a_{j1}^0)^2 = 4.190^2 + 4.297^2 + (-0.141)^2 = 36.048,$$

$$v_1^1 = a_{11}^0 + \text{sign}(a_{11}^0) \left(\sum_{j=1}^{n} (a_{j1}^0)^2 \right)^{1/2} = 4.190 + 1 \cdot \sqrt{36.048} = 10.194,$$

$$v_i^1 = a_{i1}^0,\ i = 2, \ldots, n,\ v^1 = \begin{pmatrix} 10.194 & 4.297 & -0.141 \end{pmatrix}^T,$$

$$v^{1T} v^1 = 36.048 - 4.190^2 + 10.194^2 = 122.414,$$

$$H_1^{(1)} = E - 2 \frac{v^1 v^{1T}}{v^{1T} v^1} = \begin{pmatrix} -0.698 & -0.716 & 0.023 \\ -0.716 & 0.698 & 0.010 \\ 0.023 & 0.010 & 1.000 \end{pmatrix}.$$

$$A_1^{(1)} = H_1^{(1)} A_0^{(1)} = \begin{pmatrix} -6.004 & -1.243 & 5.203 \\ 0 & 1.875 & -8.512 \\ 0 & 0.168 & 0.571 \end{pmatrix}.$$

Step $p = 2$. Let's find the Householder matrix $H_2^{(1)}$:

$$\sum_{j=2}^{n} (a_{j2}^1)^2 = 1.875^2 + 0.168^2 = 3.543,$$

$$v_1^2 = 0,\ v_2^2 = a_{22}^1 + \text{sign}(a_{22}^1) \left(\sum_{j=2}^{n} (a_{j2}^1)^2 \right)^{1/2} = 1.875 + 1 \cdot \sqrt{3.543} = 3.757,$$

$$v_i^2 = a_{i2}^1,\ i = 3, \ldots, n,\ v^2 = \begin{pmatrix} 0 & 3.757 & 0.168 \end{pmatrix}^T,$$

$$v^{2T} v^2 = 3.543 - 1.875^2 + 3.757^2 = 14.146,$$

$$H_2^{(1)} = E - 2\frac{v^2 v^{2T}}{v^{2T} v^2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0.996 & -0.089 \\ 0 & -0.089 & 0.996 \end{pmatrix}.$$

$$A_2^{(1)} = H_2^{(1)} A_1^{(1)} = \begin{pmatrix} -6.004 & -1.243 & 5.203 \\ 0 & -1.882 & 8.428 \\ 0 & 0 & 1.327 \end{pmatrix} = R^{(1)}.$$

By doing $n-1=2$ steps for $p=1,\ldots,n-1=2$, we got the desired decomposition $A^{(1)} = Q^{(1)} R^{(1)}$ of the matrix $A^{(1)}$, where $R^{(1)} = A_{n-1}^{(1)} = A_2^{(1)}$ and

$$Q^{(1)} = H_1^{(1)} H_2^{(1)} \ldots H_{n-1}^{(1)} = H_1^{(1)} H_2^{(1)} = \begin{pmatrix} -0.698 & 0.711 & 0.087 \\ -0.716 & -0.696 & -0.052 \\ 0.023 & -0.099 & 0.995 \end{pmatrix}.$$

Let's calculate the matrix $A^{(2)}$:

$$A^{(2)} = R^{(1)} Q^{(1)} = \begin{pmatrix} 5.202 & -3.917 & 4.719 \\ 1.545 & 0.478 & 8.483 \\ 0.031 & -0.131 & 1.320 \end{pmatrix}.$$

Let's calculate the error $\varepsilon^{(2)}$:

$$\varepsilon^{(2)} = \left(\sum_{l>m+1}^{n} (a_{lm}^{(2)})^2 + \min_{l=m+1}^{n} (a_{lm}^{(2)})^2 \right)^{1/2} = (0.031^2 + (-0.131)^2)^{1/2} = 0.135 > \varepsilon.$$

The following iterations $k = 3,4,5,\ldots$ are completed in a similar way:

$$R^{(2)} = \begin{pmatrix} -5.427 & 3.619 & -6.946 \\ 0 & -1.577 & -6.682 \\ 0 & 0 & 1.753 \end{pmatrix}, \; Q^{(2)} = \begin{pmatrix} -0.959 & 0.284 & -0.025 \\ -0.285 & -0.956 & 0.066 \\ -0.006 & 0.070 & 0.998 \end{pmatrix},$$

$$A^{(3)} = R^{(2)} Q^{(2)} = \begin{pmatrix} 4.211 & -5.487 & -6.553 \\ 0.487 & 1.039 & -6.769 \\ -0.010 & 0.123 & 1.749 \end{pmatrix}, \; \varepsilon^{(3)} = 0.123 > \varepsilon;$$

$$R^{(3)} = \begin{pmatrix} -4.240 & 5.332 & 7.291 \\ 0 & -1.667 & 5.844 \\ 0 & 0 & 2.123 \end{pmatrix}, \; Q^{(3)} = \begin{pmatrix} -0.993 & 0.114 & 0.010 \\ -0.115 & -0.991 & -0.065 \\ 0.002 & -0.066 & 0.998 \end{pmatrix},$$

$$A^{(4)} = R^{(3)} Q^{(3)} = \begin{pmatrix} 3.616 & -6.253 & 6.884 \\ 0.205 & 1.265 & 5.940 \\ 0.005 & -0.140 & 2.118 \end{pmatrix}, \; \varepsilon^{(4)} = 0.141 > \varepsilon;$$

$$R^{(4)} = \begin{pmatrix} -3.622 & 6.171 & -7.213 \\ 0 & -1.623 & -5.351 \\ 0 & 0 & 2.551 \end{pmatrix},\ Q^{(4)} = \begin{pmatrix} -0.998 & 0.056 & -0.006 \\ -0.057 & -0.995 & 0.081 \\ -0.001 & 0.081 & 0.997 \end{pmatrix},$$

$$A^{(5)} = R^{(4)}Q^{(4)} = \begin{pmatrix} 3.277 & -6.931 & -6.667 \\ 0.099 & 1.180 & -5.464 \\ -0.004 & 0.207 & 2.543 \end{pmatrix},\ \varepsilon^{(5)} = 0.0995 > \varepsilon;\ \text{etc.:}$$

$$A^{(6)} = R^{(5)}Q^{(5)} = \begin{pmatrix} 3.075 & -7.889 & 5.765 \\ 0.048 & 0.700 & 4.996 \\ 0.004 & -0.464 & 3.225 \end{pmatrix},\ \varepsilon^{(6)} = 0.0480 > \varepsilon;$$

$$A^{(7)} = R^{(6)}Q^{(6)} = \begin{pmatrix} 2.959 & -9.762 & -1.279 \\ 0.018 & -0.501 & -2.849 \\ -0.006 & 2.511 & 4.542 \end{pmatrix},\ \varepsilon^{(7)} = 0.0187 > \varepsilon;$$

$$A^{(8)} = R^{(7)}Q^{(7)} = \begin{pmatrix} 2.903 & -0.433 & 9.864 \\ -0.005 & 4.447 & 3.360 \\ -0.004 & -1.972 & -0.351 \end{pmatrix},\ \varepsilon^{(8)} = 0.0066 > \varepsilon;$$

$$A^{(9)} = R^{(8)}Q^{(8)} = \begin{pmatrix} 2.890 & -4.396 & -8.839 \\ -0.013 & 3.139 & -4.920 \\ 0.001 & 0.431 & 0.971 \end{pmatrix},\ \varepsilon^{(9)} = 0.0133 > \varepsilon;$$

$$A^{(10)} = R^{(9)}Q^{(9)} = \begin{pmatrix} 2.906 & -5.568 & 8.129 \\ -0.017 & 2.461 & 5.167 \\ -0.001 & -0.227 & 1.633 \end{pmatrix},\ \varepsilon^{(10)} = 0.0168 > \varepsilon;$$

$$A^{(11)} = R^{(10)}Q^{(10)} = \begin{pmatrix} 2.936 & -6.298 & -7.544 \\ -0.016 & 1.958 & -5.244 \\ 0.001 & 0.198 & 2.106 \end{pmatrix},\ \varepsilon^{(11)} = 0.0156 > \varepsilon;$$

$$A^{(12)} = R^{(11)}Q^{(11)} = \begin{pmatrix} 2.968 & -7.033 & 6.828 \\ -0.011 & 1.406 & 5.211 \\ -0.001 & -0.272 & 2.627 \end{pmatrix},\ \varepsilon^{(12)} = 0.0113 > \varepsilon;$$

$$A^{(13)} = R^{(12)}Q^{(12)} = \begin{pmatrix} 2.993 & -8.216 & -5.312 \\ -0.006 & 0.479 & -4.811 \\ 0.001 & 0.699 & 3.527 \end{pmatrix},\ \varepsilon^{(13)} = 0.0062 > \varepsilon;$$

$$A^{(14)} = R^{(13)}Q^{(13)} = \begin{pmatrix} 3.009 & -8.948 & -3.934 \\ -0.002 & 0.704 & 0.541 \\ -0.001 & -4.982 & 3.287 \end{pmatrix},\ \varepsilon^{(14)} = 0.0021 > \varepsilon;$$

$$A^{(15)} = R^{(14)}Q^{(14)} = \begin{pmatrix} 3.016 & 2.654 & 9.404 \\ -0.001 & 3.847 & -4.544 \\ 0.000 & 0.981 & 0.137 \end{pmatrix},\ \varepsilon^{(15)} = 0.0015 > \varepsilon;$$

$$A^{(16)} = R^{(15)}Q^{(15)} = \begin{pmatrix} 3.016 & 4.894 & -8.459 \\ -0.002 & 2.770 & 5.211 \\ -0.000 & -0.309 & 1.214 \end{pmatrix},\ \varepsilon^{(16)} = 0.0025 > \varepsilon;$$

$$A^{(17)} = R^{(16)}Q^{(16)} = \begin{pmatrix} 3.013 & 5.800 & 7.870 \\ -0.003 & 2.217 & -5.316 \\ 0.000 & 0.197 & 1.771 \end{pmatrix},\ \varepsilon^{(17)} = 0.0026 > \varepsilon.$$

One can see the convergence of the element $a_{11}^{(k)}$ of the matrix $A^{(k)}$, but not of the elements in the diagonal block $a_{22}^{(k)}, a_{33}^{(k)}, a_{23}^{(k)}, a_{32}^{(k)}$:

$$R^{(17)} = \begin{pmatrix} -3.013 & -5.799 & -7.874 \\ 0 & -2.230 & 5.132 \\ 0 & 0 & 2.232 \end{pmatrix},\ Q^{(17)} = \begin{pmatrix} -1.000 & -0.001 & -0.000 \\ 0.001 & -0.996 & -0.088 \\ -0.000 & -0.088 & 0.996 \end{pmatrix},$$

$$A^{(18)} = R^{(17)}Q^{(17)} = \begin{pmatrix} 3.008 & 6.474 & -7.331 \\ -0.002 & 1.768 & 5.309 \\ -0.000 & -0.197 & 2.224 \end{pmatrix},\ \varepsilon^{(18)} = 0.0021 > \varepsilon;$$

$$R^{(18)} = \begin{pmatrix} -3.008 & -6.473 & 7.335 \\ 0 & -1.784 & -5.026 \\ 0 & 0 & 2.796 \end{pmatrix},\ Q^{(18)} = \begin{pmatrix} -1.000 & -0.001 & 0.000 \\ 0.001 & -0.994 & 0.110 \\ 0.000 & 0.110 & 0.994 \end{pmatrix},$$

$$A^{(19)} = R^{(18)}Q^{(18)} = \begin{pmatrix} 3.003 & 7.245 & 6.575 \\ -0.001 & 1.218 & -5.192 \\ 0.000 & 0.309 & 2.778 \end{pmatrix},\ \varepsilon^{(19)} = 0.0014 > \varepsilon.$$

Iterative process ends on iteration $k = 20$:

$$R^{(19)} = \begin{pmatrix} -3.003 & -7.244 & -6.578 \\ 0 & -1.260 & 4.351 \\ 0 & 0 & 3.964 \end{pmatrix},\ Q^{(19)} = \begin{pmatrix} -1.000 & -0.000 & -0.000 \\ 0.000 & -0.970 & -0.245 \\ -0.000 & -0.245 & 0.970 \end{pmatrix},$$

$$A^{(20)} = R^{(19)}Q^{(19)} = \begin{pmatrix} 3.000 & 8.636 & -4.603 \\ -0.000 & 0.156 & 4.527 \\ -0.000 & -0.971 & 3.844 \end{pmatrix},\ \varepsilon^{(20)} = 0.0007 < \varepsilon.$$

The diagonal element $a_{11}^{(20)}$ of the matrix $A^{(20)}$ and a pair of complex-conjugate eigenvalues $\lambda_{2,3}$, determined from the quadratic equation

$(a_{22}^{(20)}-\lambda)(a_{33}^{(20)}-\lambda)=a_{23}^{(20)}a_{32}^{(20)}$, can be taken as the desired eigenvalues of the matrix A:

$$\lambda_1=a_{11}^{(20)},\ \lambda_{2,3}=\frac{1}{2}(a_{22}^{(20)}+a_{33}^{(20)})\pm\frac{1}{2}\sqrt{(a_{22}^{(20)}+a_{33}^{(20)})^2-4(a_{22}^{(20)}a_{33}^{(20)}-a_{23}^{(20)}a_{32}^{(20)})};$$

$$\lambda_1=3.0003,\ \lambda_2=1.9999+0.9974i,\ \lambda_3=1.9999-0.9974i.$$

2.21. PROGRAM #08

Below is a proposed variant of the program algorithm for finding the eigenvalues of an arbitrary matrix using the *QR*-decomposition method.

```
ALGORITHM     "QR-decomposition method"
INPUT         n; a[n][n], e
HELPER        i, j, p, s; f, g, v[n], h[n][n], q[n][n], b[n][n]
OUTPUT        k; w[n]
BEGIN
    ENTER n, a, e
    f:=0
    g:=a[2][1]*a[2][1]
    CYCLE "Rows" FOR i FROM 1 TO n-1 BY 1
        CYCLE "Columns" FOR j FROM i+2 TO n BY 1
            f:=f+a[j][i]*a[j][i]
        IF (a[i+1][i]*a[i+1][i]<g) g:=a[i+1][i]*a[i+1][i]
    f:=ROOT(f+g)
    k:=0
    CYCLE "Iteration" WHILE (f>e)
        #Filling the initial Q matrix with the unity matrix
        CYCLE "Rows" FOR i FROM 1 TO n BY 1
            CYCLE "Columns" FOR j FROM 1 TO n BY 1
                IF (i=j) q[i][j]:=1 ELSE q[i][j]:=0
        CYCLE "QR-decomposition" FOR p FROM 1 TO n-1 BY 1
            g:=0
            CYCLE "Rows" FOR i FROM 1 TO p-1 BY 1
```

```
            v[i]:=0
        CYCLE “Rows” FOR i FROM p TO n BY 1
            v[i]:=a[i][p]
            g:=g+a[i][p]*a[i][p]
        IF (a[p][p]>0) v[p]:=v[p]+ROOT(g)
        IF (a[p][p]<0) v[p]:=v[p]-ROOT(g)
        g:=g-a[p][p]*a[p][p]+v[p]*v[p]
        CYCLE “Rows” FOR i FROM 1 TO n BY 1
            CYCLE “Columns” FOR j FROM 1 TO n BY 1
                IF (i=j) h[i][j]:=1 ELSE h[i][j]:=0
                h[i][j]:=h[i][j]-2*v[i]*v[j]/g
        CYCLE “Rows” FOR i FROM 1 TO n BY 1
            CYCLE “Columns” FOR j FROM 1 TO n BY 1
                b[i][j]:=0
                CYCLE “Item” FOR s FROM 1 TO n BY 1
                    b[i][j]:=b[i][j]+h[i][s]*a[s][j]
        CYCLE “Rows” FOR i FROM 1 TO n BY 1
            CYCLE “Columns” FOR j FROM 1 TO n BY 1
                a[i][j]:=b[i][j]
        CYCLE “Rows” FOR i FROM 1 TO n BY 1
            CYCLE “Columns” FOR j FROM 1 TO n BY 1
                b[i][j]:=0
                CYCLE “Item” FOR s FROM 1 TO n BY 1
                    b[i][j]:=b[i][j]+q[i][s]*h[s][j]
        CYCLE “Rows” FOR i FROM 1 TO n BY 1
            CYCLE “Columns” FOR j FROM 1 TO n BY 1
                q[i][j]:=b[i][j]
#Matrices Q and R are multiplied in reverse order
CYCLE “Rows” FOR i FROM 1 TO n BY 1
    CYCLE “Columns” FOR j FROM 1 TO n BY 1
```

```
                        b[i][j]:=0
                        CYCLE “Item” FOR s FROM 1 TO n BY 1
                            b[i][j]:=b[i][j]+a[i][s]*q[s][j]
                CYCLE “Rows” FOR i FROM 1 TO n BY 1
                    CYCLE “Columns” FOR j FROM 1 TO n BY 1
                        a[i][j]:=b[i][j]
                f:=0
                g:=a[2][1]*a[2][1]
                CYCLE “Rows” FOR i FROM 1 TO n-1 BY 1
                    CYCLE “Columns” FOR j FROM i+2 TO n BY 1
                        f:=f+a[j][i]*a[j][i]
                    IF (a[i+1][i]*a[i+1][i]<g) g:=a[i+1][i]*a[i+1][i]
                f:=ROOT(f+g)
                k:=k+1
            j:=0
            CYCLE “Rows” FOR i FROM 1 TO n-1 BY 1
                IF (j=0)
                    #Checking for the presence of a complex-conjugate pair
                    IF (a[i+1][i]<=10*f)OR(-a[i+1][i]<=-10*f) w[i]:=a[i][i]
                    ELSE
                        w[i]:=(a[i][i]+a[i+1][i+1])/2+ROOT((a[i][i]+
a[i+1][i+1])*(a[i][i]+a[i+1][i+1])-4*(a[i][i]*a[i+1][i+1]-a[i][i+1]*a[i+1][i]))/2
                        w[i+1]:=(a[i][i]+a[i+1][i+1])/2-ROOT((a[i][i]+
a[i+1][i+1])*(a[i][i]+a[i+1][i+1])-4*(a[i][i]*a[i+1][i+1]-a[i][i+1]*a[i+1][i]))/2
                        j:=1
                ELSE j:=0
            IF (j=0) w[n]:=a[n][n]
            PRINT k, w
        END
```

2.22. INVERSE ITERATION METHOD

The *QR*-decomposition method described above solves the full problem of eigenvalues of arbitrary matrices. But the problem of finding the corresponding eigenvectors for known approximations of the eigenvalues remains unsolved. The inverse iteration method (or inverse power method) is intended to solve it.

Suppose we have an arbitrary matrix A and some approximate value μ of its eigenvalue λ is known, but not the corresponding eigenvector v. Then, having chosen some initial vector $y^{(0)}$, for example, the vector $y^{(0)} = (1\,1\ldots1)^T$, the components of which are equal to one, we can construct the following iterative process to find v:

$$y^{(k)} = (A - \mu E)^{-1} y^{(k-1)}.$$

To remove a drawback associated with a great change in the components of the iterated vector $y^{(k)}$ during the iterative process, usually at each step it is normalized to a unit length as follows:

$z^{(k)} = (A - \mu E)^{-1} y^{(k-1)}$, $y^{(k)} = \dfrac{z^{(k)}}{\left\| z^{(k)} \right\|}$, herewith, a vector with a unit norm is taken as an initial approximation $y^{(0)}$.

The resulting sequence of vectors $y^{(k)}$ does not necessarily converge, but starting from a certain step, any vector in the sequence is an eigenvector v with the required accuracy. It corresponds to the eigenvalue closest to μ (usually λ). Then, after the eigenvector v is found, this value can be calculated exactly using the formula

$\lambda = \dfrac{v^T A v}{v^T v}$, accordingly, $\lambda^{(k)} = \dfrac{y^{(k)T} A y^{(k)}}{y^{(k)T} y^{(k)}}$.

The following condition is used as a criterion for the termination of calculations:

$$\varepsilon^{(k)} = \left| \lambda^{(k)} - \lambda^{(k-1)} \right| \le \varepsilon,$$

where ε is the calculation accuracy set by the calculator, and $\lambda^{(k)}$ is an eigenvalue corresponding to the vector $v \approx y^{(k)}$.

The rate of convergence of the process is determined by the accuracy of the initial approximation μ: the higher it is, the faster the convergence.

It is easy to see that the inverse iteration method differs from the power method only in the matrix $(A-\mu E)^{-1}$ used for multiplication, which results in additional around n^3 operations for matrix inversion. Ultimately, it allows to find the eigenvector corresponding to the maximum in modulus eigenvalue of the matrix $(A-\mu E)^{-1}$.

It's not hard to show that the eigenvectors of the matrices A and $(A-\mu E)^{-1}$ coincide: $Av=\lambda v$, $Av-\mu v=\lambda v-\mu v$, $(A-\mu E)v=(\lambda-\mu)v$, $(\lambda-\mu)^{-1}v=(A-\mu E)^{-1}v$.

Herewith, the eigenvalues of the matrix $(A-\mu E)^{-1}$ are equal to $(\lambda_i-\mu)^{-1}$, where λ_i are the eigenvalues of the matrix A. And then the found maximum in modulus eigenvalue $(\lambda_i-\mu)^{-1}$ corresponds to the minimum in modulus value $(\lambda_i-\mu)$, which is what was required in the given problem.

Example. Calculate the eigenvectors of the matrix $A=\begin{pmatrix}4 & -1 & -5\\ 2 & 5 & 8\\ 1 & -1 & -2\end{pmatrix}$ using the inverse iteration method (with normalization of the iterated vector) with an accuracy of $\varepsilon=0.2$, refine the already known eigenvalues of the matrix A obtained by the *QR*-decomposition method:

$\lambda_1=3.0003,\ \lambda_2=1.9999+0.9974i,\ \lambda_3=1.9999-0.9974i$.

To find the corresponding eigenvectors of the matrix A, let's use the power method for three matrices $B_i=(A-\lambda_i E)^{-1}$, which can be calculated, for example, using the Gaussian method for matrix inversion.

$$B_1=(A-\lambda_1 E)^{-1}=\begin{pmatrix}3335.19 & 0.50 & -3334.19\\ -30031.19 & -2.00 & 30026.19\\ 6672.88 & 0.50 & -6671.88\end{pmatrix},$$

$$B_2=(A-\lambda_2 E)^{-1}=\begin{pmatrix}-555.53-417.62i & 192.41+11.10i & 1143.86+259.01i\\ 1251.48+1809.39i & -566.13-225.21i & -3175.59-1920.39i\\ -556.03-418.12i & 192.41+11.10i & 1144.37+259.51i\end{pmatrix},$$

$$B_3 = (A - \lambda_3 E)^{-1} = \begin{pmatrix} -555.53 + 417.62i & 192.41 - 11.10i & 1143.86 - 259.01i \\ 1251.48 - 1809.39i & -566.13 + 225.21i & -3175.59 + 1920.39i \\ -556.03 + 418.12i & 192.41 - 11.10i & 1144.37 - 259.51i \end{pmatrix}.$$

Let's take $y^{(0)} = (1 \quad 1 \quad 1)^T$ as the initial approximation of the eigenvector. Let's implement the iterative process of the inverse iteration method using the norm $\|\cdot\|_c$ and iteration termination condition $\left|\lambda^{(k)} - \lambda^{(k-1)}\right| \le \varepsilon$:

$$z_i^{(k)} = B_i y_i^{(k-1)}, \quad y_i^{(k)} = \frac{z_i^{(k)}}{\left\|z_i^{(k)}\right\|_c}, \quad \lambda_i^{(k)} = \frac{y_i^{(k)T} A y_i^{(k)}}{y_i^{(k)T} y_i^{(k)}}.$$

For λ_1 the iterative process is done in the following way:

$$z_1^{(1)} = B_1 y^{(0)} = \left(1.4994 \quad -6.9981 \quad 1.4994\right)^T,$$

$$y_1^{(1)} = \frac{z_1^{(1)}}{\left\|z_1^{(1)}\right\|_c} = \left(0.2143 \quad -1.0000 \quad 0.2143\right)^T,$$

$$\lambda_1^{(1)} = \frac{y_1^{(1)T} A y_1^{(1)}}{y_1^{(1)T} y_1^{(1)}} = 2.9255;$$

$$z_1^{(2)} = B_1 y_1^{(1)} = \left(-0.2857 \quad 0.9286 \quad -0.2857\right)^T,$$

$$y_1^{(2)} = \frac{z_1^{(2)}}{\left\|z_1^{(2)}\right\|_c} = \left(-0.3077 \quad 1.0000 \quad -0.3077\right)^T,$$

$$\lambda_1^{(2)} = \frac{y_1^{(2)T} A y_1^{(2)}}{y_1^{(2)T} y_1^{(2)}} = 1.9754, \ \varepsilon_1^{(2)} = \left|\lambda_1^{(2)} - \lambda_1^{(1)}\right| = 0.9501 > \varepsilon;$$

$$z_1^{(3)} = B_1 y_1^{(2)} = \left(0.1923 \quad -0.4617 \quad 0.1923\right)^T,$$

$$y_1^{(3)} = \frac{z_1^{(3)}}{\left\|z_1^{(3)}\right\|_c} = \left(0.4166 \quad -1.0000 \quad 0.4166\right)^T,$$

$$\lambda_1^{(3)} = \frac{y_1^{(3)T} A y_1^{(3)}}{y_1^{(3)T} y_1^{(3)}} = 0.9803, \ \varepsilon_1^{(3)} = \left|\lambda_1^{(3)} - \lambda_1^{(2)}\right| = 0.9951 > \varepsilon; \text{ etc.}$$

For λ_1 the iterative process stops at iteration $k = 94$:

$$z_1^{(92)} = B_1 y_1^{(91)} = \left(8.5527 \quad -77.8404 \quad 17.1940\right)^T,$$

$$y_1^{(92)} = \frac{z_1^{(92)}}{\left\|z_1^{(92)}\right\|_c} = \left(0.1099 \quad -1.0000 \quad 0.2209\right)^T,$$

$$\lambda_1^{(92)} = \frac{y_1^{(92)T} A y_1^{(92)}}{y_1^{(92)T} y_1^{(92)}} = 3.0141, \ \varepsilon_1^{(92)} = \left|\lambda_1^{(92)} - \lambda_1^{(91)}\right| = 2.0458 > \varepsilon;$$

$$z_1^{(93)} = B_1 y_1^{(92)} = \left(-370.53 \quad 3334.74 \quad -741.05\right)^T,$$

$$y_1^{(93)} = \frac{z_1^{(93)}}{\left\| z_1^{(93)} \right\|_c} = \left(-0.1111 \quad 1.0000 \quad -0.2222\right)^T,$$

$$\lambda_1^{(93)} = \frac{y_1^{(93)T} A y_1^{(93)}}{y_1^{(93)T} y_1^{(93)}} = 3.0000,\ \varepsilon_1^{(93)} = \left|\lambda_1^{(93)} - \lambda_1^{(92)}\right| = 0.0141 > \varepsilon;$$

$$z_1^{(94)} = B_1 y_1^{(93)} = \left(370.85 \quad -3337.68 \quad 741.71\right)^T,$$

$$y_1^{(94)} = \frac{z_1^{(94)}}{\left\| z_1^{(94)} \right\|_c} = \left(0.1111 \quad -1.0000 \quad 0.2222\right)^T,$$

$$\lambda_1^{(94)} = \frac{y_1^{(94)T} A y_1^{(94)}}{y_1^{(94)T} y_1^{(94)}} = 3.0000,\ \varepsilon_1^{(94)} = \left|\lambda_1^{(94)} - \lambda_1^{(93)}\right| = 1.40 \cdot 10^{-7} < \varepsilon.$$

For λ_2 the iterative process stops at iteration $k = 3$:

$$z_2^{(1)} = B_2 y^{(0)} = \left(780.74 - 147.51i \quad -2490.24 - 336.21i \quad 780.74 - 147.51i\right)^T,$$

$$y_2^{(1)} = \frac{z_2^{(1)}}{\left\| z_2^{(1)} \right\|_c} = \left(0.3107 - 0.0587i \quad -0.9910 - 0.1338i \quad 0.3107 - 0.0587i\right)^T,$$

$$\lambda_2^{(1)} = \frac{y_2^{(1)T} A y_2^{(1)}}{y_2^{(1)T} y_2^{(1)}} = 1.9997 + 0.9975i;$$

$$z_2^{(2)} = B_2 y_2^{(1)} = \left(-15.71 - 120.56i \quad -73.43 + 377.39i \quad -15.71 - 120.56i\right)^T,$$

$$y_2^{(2)} = \frac{z_2^{(2)}}{\left\| z_2^{(2)} \right\|_c} = \left(-0.0409 - 0.3136i \quad -0.1910 + 0.9816i \quad -0.0409 - 0.3136i\right)^T,$$

$$\lambda_2^{(2)} = \frac{y_2^{(2)T} A y_2^{(2)}}{y_2^{(2)T} y_2^{(2)}} = 2.0000 + 1.0000i,\ \varepsilon_2^{(2)} = \left|\lambda_2^{(2)} - \lambda_2^{(1)}\right| = 0.0025 > \varepsilon;$$

$$z_2^{(3)} = B_2 y_2^{(2)} = \left(-121.42 + 8.74i \quad 373.01 + 95.19i \quad -121.42 + 8.74i\right)^T,$$

$$y_2^{(3)} = \frac{z_2^{(3)}}{\left\| z_2^{(3)} \right\|_c} = \left(-0.3154 + 0.0227i \quad 0.9689 + 0.2473i \quad -0.3154 + 0.0227i\right)^T,$$

$$\lambda_2^{(3)} = \frac{y_2^{(3)T} A y_2^{(3)}}{y_2^{(3)T} y_2^{(3)}} = 2.0000 + 1.0000i,\ \varepsilon_2^{(3)} = \left|\lambda_2^{(3)} - \lambda_2^{(2)}\right| = 3.29 \cdot 10^{-6} < \varepsilon.$$

For λ_3 the iterative process stops at iteration $k = 3$:

$$z_3^{(1)} = B_3 y^{(0)} = \left(780.74 + 147.51i \quad -2490.24 + 336.21i \quad 780.74 + 147.51i\right)^T,$$

$$y_3^{(1)} = \frac{z_3^{(1)}}{\left\| z_3^{(1)} \right\|_c} = \left(0.3107 + 0.0587i \quad -0.9910 + 0.1338i \quad 0.3107 + 0.0587i\right)^T,$$

$$\lambda_3^{(1)} = \frac{y_3^{(1)T} A y_3^{(1)}}{y_3^{(1)T} y_3^{(1)}} = 1.9997 - 0.9975i;$$

$$z_3^{(2)} = B_3 y_3^{(1)} = \left(-15.71 + 120.56i \quad -73.43 - 377.39i \quad -15.71 + 120.56i\right)^T,$$

$$y_3^{(2)} = \frac{z_3^{(2)}}{\left\| z_3^{(2)} \right\|_c} = \left(-0.0409 + 0.3136i \quad -0.1910 - 0.9816i \quad -0.0409 + 0.3136i\right)^T,$$

$$\lambda_3^{(2)} = \frac{y_3^{(2)T} A y_3^{(2)}}{y_3^{(2)T} y_3^{(2)}} = 2.0000 - 1.0000i,\ \varepsilon_3^{(2)} = \left|\lambda_3^{(2)} - \lambda_3^{(1)}\right| = 0.0025 > \varepsilon;$$

$$z_3^{(3)} = B_3 y_3^{(2)} = \left(-121.42 - 8.74i \quad 373.01 - 95.19i \quad -121.42 - 8.74i\right)^T,$$

$$y_3^{(3)} = \frac{z_3^{(3)}}{\left\| z_3^{(3)} \right\|_c} = \left(-0.3154 - 0.0227i \quad 0.9689 - 0.2473i \quad -0.3154 - 0.0227i\right)^T,$$

$$\lambda_3^{(3)} = \frac{y_3^{(3)T} A y_3^{(3)}}{y_3^{(3)T} y_3^{(3)}} = 2.0000 - 1.0000i,\ \varepsilon_3^{(3)} = \left|\lambda_3^{(3)} - \lambda_3^{(2)}\right| = 3.29 \cdot 10^{-6} < \varepsilon.$$

Thus, the eigenvalues and eigenvectors of the matrix A obtained by the inverse iteration method are:

$$\lambda_1 = 3.0000,\ \lambda_2 = 2.0000 + 1.0000i,\ \lambda_3 = 2.0000 - 1.0000i;$$

$$v^1 = \begin{pmatrix} 0.1111 \\ -1.0000 \\ 0.2222 \end{pmatrix},\ v^2 = \begin{pmatrix} -0.3154 + 0.0227i \\ 0.9689 + 0.2473i \\ -0.3154 + 0.0227i \end{pmatrix},\ v^3 = \begin{pmatrix} -0.3154 - 0.0227i \\ 0.9689 - 0.2473i \\ -0.3154 - 0.0227i \end{pmatrix}.$$

It is not hard to check that all the obtained eigenvalues and eigenvectors satisfy the equation $Ax = \lambda x$ within the specified accuracy:

$$Av_1 - \lambda_1 v_1 \approx 0,\ Av_2 - \lambda_2 v_2 \approx 0,\ Av_3 - \lambda_3 v_3 \approx 0.$$

2.23. PROGRAM #09

Below is a proposed variant of the program algorithm for finding the eigenvector of a matrix and refining the corresponding eigenvalue using the inverse iteration method.

ALGORITHM	“Inverse iteration method”
INPUT	n; a[n][n], b[n][n], e

```
HELPER          i, j; f, r, z[n]
OUTPUT          k; w, y[n]
BEGIN
    ENTER n, a, b, e
    #Filling the initial eigenvector
    CYCLE “Rows” FOR i FROM 1 TO n BY 1
        y[i]:=1
    f:=2*e
    k:=0
    CYCLE “Iteration” WHILE (f>e)
        r:=0
        CYCLE “Rows” FOR i FROM 1 TO n BY 1
            z[i]:=0
            CYCLE “Item” FOR j FROM 1 TO n BY 1
                z[i]:=z[i]+b[i][j]*y[j]
            IF (z[i]>r) r:=z[i]
            IF (-z[i]>r) r:=-z[i]
        IF (k>0) f:=w
        CYCLE “Rows” FOR i FROM 1 TO n BY 1
            y[i]:=z[i]/r
        #Refining the eigenvalue
        CYCLE “Rows” FOR i FROM 1 TO n BY 1
            z[i]:=0
            CYCLE “Item” FOR j FROM 1 TO n BY 1
                z[i]:=z[i]+a[i][j]*y[j]
        w:=0
        r:=0
        CYCLE “Item” FOR j FROM 1 TO n BY 1
            w:=w+y[j]*z[j]
            r:=r+y[j]*y[j]
```

```
            w:=w/r
            IF (k>0) f:=w-f
            IF (f<0) f:=-f
            k:=k+1
        PRINT k, w, y
END
```